

Question ID: 98c12e38

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Area and volume	Medium

Question

Circle K has a radius of 4 millimeters (mm). Circle L has an area of $100\pi \text{ mm}^2$. What is the total area, in mm^2 , of circles K and L ?

Answer

- A. 14π
- B. 28π
- C. 56π
- D. 116π

Correct Answer: D

Rationale

Choice D is correct. The area, A , of a circle is given by the formula $A = \pi r^2$, where r represents the radius of the circle. It's given that circle K has a radius of 4 millimeters (mm). Substituting 4 for r in the formula $A = \pi r^2$ yields $A = \pi(4)^2$, or $A = 16\pi$. Therefore, the area of circle K is $16\pi \text{ mm}^2$. It's given that circle L has an area of $100\pi \text{ mm}^2$. Therefore, the total area, in mm^2 , of circles K and L is $16\pi + 100\pi$, or 116π .

Choice A is incorrect. This is the sum of the radii, in mm, of circles K and L multiplied by π , not the total area, in mm^2 , of the circles.

Choice B is incorrect. This is the sum of the diameters, in mm, of circles K and L multiplied by π , not the total area, in mm^2 , of the circles.

Choice C is incorrect and may result from conceptual or calculation errors.